A Technical Introduction to Linear discriminant analysis

Section 1

This means that the criterion of an input x → {\displaystyle {\vec {x}}} being in a class y {\displaystyle y} is purely a function of this linear combination of the known observations. It is often useful to see this conclusion in geometrical terms: the criterion of an input x → {\displaystyle {\vec {x}}} being in a class y {\displaystyle y} is purely a function of projection of multidimensional-space point x → {\displaystyle {\vec {x}}} onto vector w → {\displaystyle {\vec {w}}} (thus, we only consider its direction). In other words, the observation belongs to y {\displaystyle y} if corresponding x → {\displaystyle {\vec {x}}} is located on a certain side of a hyperplane perpendicular to w → {\displaystyle {\vec {w}}} . The location of the plane is defined by the threshold c {\displaystyle c} .

Section 2

Be sure to note that the vector w → {\displaystyle {\vec {w}}} is the normal to the discriminant hyperplane. As an example, in a two dimensional problem, the line that best divides the two groups is perpendicular to w → {\displaystyle {\vec {w}}} . Generally, the data points to be discriminated are projected onto w → {\displaystyle {\vec {w}}} ; then the threshold that best separates the data is chosen from analysis of the one-dimensional distribution. There is no general rule for the threshold. However, if projections of points from both classes exhibit approximately the same distributions, a good choice would be the hyperplane between projections of the two means, w → · μ → 0 {\displaystyle {\vec {w}}\cdot {\vec {\mu }}_{0}} and w → · μ → 1 {\displaystyle {\vec {w}}\cdot {\vec {\mu }}_{1}} . In this case the parameter c in threshold condition w → · x → > c {\displaystyle {\vec {w}}\cdot {\vec {x}}>c} can be found explicitly:

Section 3

Earth science This method can be used to separate the alteration zones. For example, when different data from various zones are available, discriminant analysis can find the pattern within the data and classify it effectively.

Section 4

Discriminant Correlation Analysis (DCA) of the Haghighat article (see above) ALGLIB contains open-source LDA implementation in C# / C++ / Pascal / VBA. LDA in Python- LDA implementation in Python LDA tutorial using MS Excel Biomedical statistics. Discriminant analysis StatQuest: Linear Discriminant Analysis (LDA) clearly explained on YouTube Course notes, Discriminant function analysis by G. David Garson, NC State University Discriminant analysis tutorial in Microsoft Excel by Kardi Teknomo Course notes, Discriminant function analysis by David W. Stockburger, Missouri State University Archived 2016-03-03 at the Wayback Machine Discriminant function analysis (DA) by John Poulsen and Aaron French, San Francisco State University Archived 2011-12-15 at the Wayback Machine

Section 5

where I {\displaystyle I} is the identity matrix, and λ {\displaystyle \lambda } is the shrinkage intensity or regularisation parameter. This leads to the framework of regularized discriminant analysis or shrinkage discriminant analysis. Also, in many practical cases linear discriminants are not suitable. LDA and Fisher's discriminant can be extended for use in non-linear classification via the kernel trick. Here, the original observations are effectively mapped into a higher dimensional non-linear space. Linear classification in this non-linear space is then equivalent to non-linear classification in the original space. The most commonly used example of this is the kernel Fisher discriminant. LDA can be generalized to multiple discriminant analysis, where c becomes a categorical variable with N possible states, instead of only two. Analogously, if the class-conditional densities p (x → ■ c = i) {\displaystyle p({\vec {x}}\mid c=i)} are normal with shared covariances, the sufficient statistic for P (c ■ x →) {\displaystyle P(c\mid {\vec {x}})} are the values of N projections, which are the subspace spanned by the N means, affine projected by the inverse covariance matrix. These projections can be found by solving a generalized eigenvalue problem, where the numerator is the covariance matrix formed by treating the means as the samples, and the denominator is the shared covariance matrix. See "Multiclass LDA" above for details.

Section 6

Practical use In practice, the class means and covariances are not known. They can, however, be estimated from the training set. Either the maximum likelihood estimate or the maximum a posteriori estimate may be used in place of the exact value in the above equations. Although the estimates of the covariance may be considered optimal in some sense, this does not mean that the resulting discriminant obtained by substituting these values is optimal in any sense, even if the assumption of normally distributed classes is correct. Another complication in applying LDA and Fisher's discriminant to real data occurs when the number of measurements of each sample (i.e., the dimensionality of each data vector) exceeds the number of samples in each class. In this case, the covariance estimates do not have full rank, and so cannot be inverted. There are a number of ways to deal with this. One is to use a pseudo inverse instead of the usual matrix inverse in the above formulae. However, better numeric stability may be achieved by first projecting the problem onto the subspace spanned by Σ b {\displaystyle \Sigma _{b}} . Another strategy to deal with small sample size is to use a shrinkage estimator of the covariance matrix, which can be expressed mathematically as

Section 7

x → T Σ 0 – 1 x → = x → T Σ 1 – 1 x → {\displaystyle {\vec {x}}^{\mathrm {T} }\Sigma _{0}^{-1}{\vec {x}}={\vec {x}}^{\mathrm {T} }\Sigma _{1}^{-1}{\vec {x}}}

Section 8

Formulate the problem and gather data—Identify the salient attributes consumers use to evaluate products in this category—Use quantitative marketing research techniques (such as surveys) to collect data from a sample of potential customers concerning their ratings of all the product attributes. The data collection stage is usually done by marketing research professionals. Survey questions ask the respondent to rate a product from one to five (or 1 to 7, or 1 to 10) on a range of attributes chosen by the researcher. Anywhere from five to twenty attributes are chosen. They could include things like: ease of use, weight, accuracy, durability, colourfulness, price, or size. The attributes chosen will vary depending on the product being studied. The same question is asked about all the products in the study. The data for multiple products is codified and input into a statistical program such as R, SPSS or SAS. (This step is the same as in Factor analysis). Estimate the Discriminant Function Coefficients and determine the statistical significance and validity—Choose the appropriate discriminant analysis method. The direct method involves estimating the discriminant function so that all the predictors are assessed simultaneously. The stepwise method enters the predictors sequentially. The two-group method should be used when the dependent variable has two categories or states. The multiple discriminant method is used when the dependent variable has three or more categorical states. Use Wilks's Lambda to test for significance in SPSS or F stat in SAS. The most common method used to test validity is to split the sample into an estimation or analysis sample, and a validation or holdout sample. The estimation sample is used in constructing the discriminant function. The validation sample is used to construct a classification matrix which contains the number of correctly classified and incorrectly classified cases. The percentage of correctly classified cases is called the hit ratio. Plot the results on a two dimensional map, define the dimensions, and interpret the results. The statistical program (or a related module) will map the results. The map will plot each product (usually in two-dimensional space). The distance of products to each other indicate either how different they are. The dimensions must be labelled by the researcher. This requires subjective judgement and is often very challenging. See perceptual mapping.